

INGREDIENTS

① *Jacobian term:*

transformation of coordinates:

$$\mathcal{L}(\varphi, \partial_m \varphi, x) = \left| \frac{\partial x'}{\partial x} \right| \mathcal{L}(\varphi', \partial'_m \varphi', x')$$

$$x'^m = x^m + \delta x^m$$

absolute value of the determinant
of the Jacobian matrix

Hence,

$$\frac{\partial x'^m}{\partial x^p} = \frac{\partial x^m}{\partial x^p} + \frac{\partial \delta x^m}{\partial x^p} = \underbrace{\delta_p^m}_{\text{zeroth order}} + \underbrace{\partial_p \delta x^m}_{\text{1st order in "}\delta x\text{"}}, \quad \partial_p = \frac{\partial}{\partial x^p}.$$

$$\frac{\partial x'^m}{\partial x^p} = \delta_p^m + \partial_p \delta x^m$$

$\underbrace{\delta_p^m}_{\text{zeroth order}} \quad \underbrace{\partial_p \delta x^m}_{\text{1st order in "}\delta x\text{"}}$

$\Rightarrow$ Jacobian matrix has the structure $\mathbf{1} + \varepsilon$ ($\varepsilon \ll 1$).

$$\det(\mathbf{1} + \varepsilon) = \exp[\text{Tr} \ln(\mathbf{1} + \varepsilon)]$$

$$\text{1st order} \simeq \exp[\text{Tr}(\varepsilon)], \quad \text{1st order} = 1 + \text{Tr}(\varepsilon).$$

$$\therefore \det\left(\frac{\partial x'}{\partial x}\right) = 1 + \partial_m \delta x^m \quad \text{Einstein summation convention (one repeated index)}$$

But since $\partial_m \delta x^m$ is a first order correction, then its absolute value is smaller than one

$$\Rightarrow \left| \det\left(\frac{\partial x'}{\partial x}\right) \right| = \det\left(\frac{\partial x'}{\partial x}\right) = 1 + \partial_m \delta x^m.$$

② the $\varphi'(x')$ term:

we follow that:

$$\varphi'(x') = \varphi'(x + \delta x) \quad \text{coordinate transformation.}$$

$$\begin{array}{l} \text{Expansion} \\ \text{at first order} \end{array} \Rightarrow \varphi'(x') = \underbrace{\varphi'(x)}_{\text{second order in "}\delta\text{"}} + \sum_M \left(\frac{\partial \varphi'}{\partial x'^M} \right)_{x'=x} \delta x^M.$$

The derivative can be approximated by a zeroth order term coming from another expansion

① δx^M

1st order term in "δx"

$$\text{But:} \quad \left(\frac{\partial \varphi'}{\partial x'^m} \right) = \underbrace{\left(\frac{\partial \varphi}{\partial x^m} \right)}_{\text{second order terms}} + \text{1st order correction terms in "}\delta\text{"}.$$

$$\therefore \varphi'(x') = \varphi'(x) + \sum_m (\partial_m \varphi) \delta x^m.$$

Moreover:

**** Definition:** we introduce the variation of the fields $\delta\varphi(x)$ **AT THE SAME POINT** x by:

$$\delta\varphi(x) = \varphi'(x) - \varphi(x).$$

Finally:

$$\boxed{\varphi'(x') = \varphi(x) + \delta\varphi(x) + (\partial_m \varphi) \delta x^m}.$$

③ the $(\partial_m \varphi')(x')$ term:

Starting point:

$$\begin{aligned} \frac{\partial \varphi'}{\partial x'^m} &= \frac{\partial}{\partial x'^m} \{ \varphi(x) + \delta\varphi(x) + (\partial_p \varphi) \delta x^p \} \\ &= \frac{\partial x^n}{\partial x'^m} \frac{\partial}{\partial x^n} \{ \varphi(x) + \delta\varphi(x) + (\partial_p \varphi) \delta x^p \} \\ &= (\delta_m^n - \partial_m \delta x^n) \{ \partial_n \varphi + \partial_n \delta\varphi + \partial_n [(\partial_p \varphi) \delta x^p] \} \\ &= (\delta_m^n - \partial_m \delta x^n) \{ \partial_n \varphi + \partial_n \delta\varphi + (\partial_n \partial_p \varphi) \delta x^p + (\partial_p \varphi) (\partial_n \delta x^p) \}. \end{aligned}$$

inverse of the Jacobian matrix:

$$\frac{\partial x'^m}{\partial x^n} = \delta_n^m + \partial_n \delta x^m, \quad (\mathbf{1} + \varepsilon)^{-1} \simeq \mathbf{1} - \varepsilon \quad \implies \quad \frac{\partial x^n}{\partial x'^m} = \delta_m^n - \partial_m \delta x^n.$$

$$\begin{aligned} \frac{\partial \varphi'}{\partial x'^m} &= (\delta_m^n - \partial_m \delta x^n) \left\{ \underbrace{\partial_n \varphi}_{\text{0th order}} + \underbrace{\partial_n \delta \varphi}_{\text{second order}} + \underbrace{(\partial_n \partial_p \varphi) \delta x^p}_{\text{first order}} + (\partial_p \varphi)(\partial_n \delta x^p) \right\} \\ &= \partial_m \varphi + \partial_m \delta \varphi + (\partial_m \partial_p \varphi) \delta x^p + (\partial_p \varphi)(\partial_m \delta x^p) - (\partial_m \delta x^n)(\partial_n \varphi). \end{aligned}$$

~~the two coordinate terms cancel~~ $- (\partial_m \delta x^n)$ contribution

$$\boxed{\partial_m \varphi' = \partial_m \varphi + \partial_m \delta \varphi + (\partial_m \partial_p \varphi) \delta x^p}.$$

④ The $\mathcal{L}(\varphi', \partial \varphi'; x')$ term:

the starting point is:

$$\begin{aligned} &\mathcal{L} \left(\underbrace{\varphi + \delta \varphi + (\partial_m \varphi) \delta x^m}_{\varepsilon}, \underbrace{\partial_m \varphi + \partial_m \delta \varphi + (\partial_m \partial_p \varphi) \delta x^p}_{\text{1st order "}\varepsilon\text{"}}, \underbrace{x^m + \delta x^m}_{\varepsilon \text{ term}} \right) \\ &\simeq \mathcal{L}(\varphi, \partial_m \varphi; x) + (\delta \varphi + (\partial_m \varphi) \delta x^m) \frac{\partial \mathcal{L}}{\partial \varphi} \\ &\quad + (\partial_m \delta \varphi + (\partial_m \partial_p \varphi) \delta x^p) \frac{\partial \mathcal{L}}{\partial (\partial_m \varphi)} + \delta x^m \frac{\partial \mathcal{L}}{\partial x^m} + \text{second order terms.} \end{aligned}$$

However, the total derivative of the Lagrangian density reads:

$$\partial_p \mathcal{L} = (\partial_p \varphi) \frac{\partial \mathcal{L}}{\partial \varphi} + (\partial_p \partial_m \varphi) \frac{\partial \mathcal{L}}{\partial (\partial_m \varphi)} + \frac{\partial \mathcal{L}}{\partial x^p}.$$

$$p \rightarrow m \quad m \rightarrow p$$

$$\begin{aligned} \text{Then:} \quad \mathcal{L}(\varphi', \partial' \varphi'; x') &= \mathcal{L}(\varphi, \partial \varphi; x) + \delta x^m \left[(\partial_m \varphi) \frac{\partial \mathcal{L}}{\partial \varphi} + (\partial_p \partial_m \varphi) \frac{\partial \mathcal{L}}{\partial (\partial_p \varphi)} + \frac{\partial \mathcal{L}}{\partial x^m} \right] \\ &\quad + \delta \varphi \frac{\partial \mathcal{L}}{\partial \varphi} + (\partial_m \delta \varphi) \frac{\partial \mathcal{L}}{\partial (\partial_m \varphi)}. \end{aligned}$$

We have found:

$$\mathcal{L}(\varphi', \partial' \varphi'; x') = \mathcal{L}(\varphi, \partial \varphi; x) + (\partial_m \mathcal{L}) \delta x^m + \delta \varphi \frac{\partial \mathcal{L}}{\partial \varphi} + (\partial_m \delta \varphi) \frac{\partial \mathcal{L}}{\partial (\partial_m \varphi)}.$$

$$\partial_m \mathcal{L}$$

$$\boxed{\text{Symmetry Condition}}$$

$$\begin{aligned} \mathcal{L}(\varphi, \partial \varphi; x) &= \left| \frac{\partial x'}{\partial x} \right| \mathcal{L}(\varphi', \partial' \varphi'; x') \\ &= (1 + \partial_m \delta x^m) \left[\mathcal{L}(\varphi, \partial \varphi; x) + (\partial_p \mathcal{L}) \delta x^p + \delta \varphi \frac{\partial \mathcal{L}}{\partial \varphi} + (\partial_p \delta \varphi) \frac{\partial \mathcal{L}}{\partial (\partial_p \varphi)} \right] \\ \implies \quad \mathcal{L} + \delta \varphi \frac{\partial \mathcal{L}}{\partial \varphi} + (\partial_p \delta \varphi) \frac{\partial \mathcal{L}}{\partial (\partial_p \varphi)} &+ (\partial_p \mathcal{L}) \delta x^p + (\partial_m \delta x^m) \mathcal{L}, \end{aligned}$$

$$(\partial_p \mathcal{L}) \delta x^p \text{ comes from "}\delta x^p\text{"} \quad (\partial_m \delta x^m) \mathcal{L} \text{ comes from "}\partial_m \delta x^m\text{"}$$

$$\boxed{\delta \mathcal{L} = \delta \varphi \frac{\partial \mathcal{L}}{\partial \varphi} + (\partial_p \delta \varphi) \frac{\partial \mathcal{L}}{\partial (\partial_p \varphi)} + \partial_p (\mathcal{L} \delta x^p)} \quad A = 0.$$

But:

$$(\partial_p \delta \varphi) \frac{\partial \mathcal{L}}{\partial (\partial_p \varphi)} = \partial_p \left[\delta \varphi \frac{\partial \mathcal{L}}{\partial (\partial_p \varphi)} \right] - \delta \varphi \partial_p \frac{\partial \mathcal{L}}{\partial (\partial_p \varphi)}.$$

$$u'v - (uv)' = -uv'$$

Then:

$$A = \delta\varphi \frac{\partial \mathcal{L}}{\partial \varphi} - \delta\varphi \partial_p \frac{\partial \mathcal{L}}{\partial (\partial_p \varphi)} + \partial_p \left(\delta\varphi \frac{\partial \mathcal{L}}{\partial (\partial_p \varphi)} \right) + \partial_p (\mathcal{L} \delta x^p) = 0.$$

$$\iff \delta\varphi \left[\frac{\partial \mathcal{L}}{\partial \varphi} - \partial_p \frac{\partial \mathcal{L}}{\partial (\partial_p \varphi)} \right] = -\partial_p \left(\mathcal{L} \delta x^p + \delta\varphi \frac{\partial \mathcal{L}}{\partial (\partial_p \varphi)} \right).$$

involves the coordinate transformations involves the field modification at the same point

• c) the Noether theorem:

Using the same assumptions as before. Moreover, $\varphi(x)$ is a solution of the equation of motion, so it satisfies

$$\frac{\partial \mathcal{L}}{\partial \varphi} - \partial_p \frac{\partial \mathcal{L}}{\partial (\partial_p \varphi)} = 0$$

$$\implies \partial_p \left(\mathcal{L} \delta x^p + \delta\varphi \frac{\partial \mathcal{L}}{\partial (\partial_p \varphi)} \right) = 0.$$

Noether Theorem: For any trajectory of a system having a symmetry, there is a current J^m which is conserved:

$$\partial_m J^m = 0$$

with

$$J^m = \mathcal{L} \delta x^m + \delta\varphi \frac{\partial \mathcal{L}}{\partial (\partial_m \varphi)}.$$

J^m : Noether current

Comments:

① properties of the J^m components.

$$\begin{cases} J^0 & \text{temporal component} \\ J^i & \text{spatial components, } i \in [1, D-1] \end{cases}$$

$$\boxed{\partial_0 J^0 + \partial_i J^i = 0} \quad \text{equation of conservation} \quad " \partial_t \rho + \text{div } \vec{J} = 0 "$$

Def: the charge Q associated to the Noether current is:

$$Q = \int_{\tilde{\Omega}_{D-1}} d^{D-1}x J^0(x) \quad \text{integration over all of the space (NOT the time).}$$

Q should in principal depend on time, however:

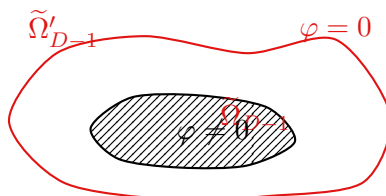
$$\frac{dQ}{dt} = \int_{\tilde{\Omega}_{D-1}} d^{D-1}x \partial_0 J^0|_{c=1} = - \int_{\tilde{\Omega}_{D-1}} d^{D-1}x \partial_i J^i = - \oint_{\partial \tilde{\Omega}_{D-1}} J^i d\Sigma_i.$$

Ostrogradsky-Green theorem $\Rightarrow$ flux integral over the boundary of $\tilde{\Omega}_{D-1}$

In physics, there are 2 possibilities:

① $\varphi \neq 0$ on a closed domain of space

$\implies$ we can choose the boundary to make the integral vanish.



$$\int_{\Omega_{D-1}} \partial_i J^i = \int_{\tilde{\Omega}_{D-1}} \partial_i J^i.$$

$$\textcircled{2} \quad \varphi \text{ decreases "fast enough" at infinity so that} \quad \frac{\partial Q}{\partial t} = 0 \quad \implies \quad Q \text{ is a constant.}$$

② Unicity of the Noether current:

The initial relation defines a structure for J^m .

But J^m is defined from its divergence. So, show that:

$$\tilde{J}^M = J^M + \partial_N V^{NM} \quad \text{where} \quad V^{NM} = -V^{MN}$$

is also a Noether current (it satisfies the conservation law).

③ how to determine the Noether current: 2 ways

1st : use the expansion of J^M

2nd : compute $\frac{\partial \mathcal{L}}{\partial \varphi} - \partial_p \frac{\partial \mathcal{L}}{\partial (\partial_p \varphi)}$ and see if it can be put into the form of a divergence term.